Aircraft Maintenance Check Scheduling (AMCS)

Business Requirement Document

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Purpose

This document covers business requirements for *aircraft maintenance check scheduling* problem. It describes operational objectives, constraints, and uncertainties that airlines face when planning maintenance activities across their fleet. The planning horizon typically spans 3-4 years for long-term capacity planning, though scheduling decisions can be made on shorter time scales (monthly, weekly, or daily) depending on the use case.

Problem Context

Aircraft maintenance is mandatory for airworthiness of a fleet and represents 9-10% of an airline's total operational costs (approximately $2.5M per aircraft per year[1]). Airlines must schedule maintenance checks before certain thresholds are reached. These thresholds are measured in:

- **Calendar Days (DY)**: 24-hour periods
- **Flight Hours (FH)**: Time between wheel lift-off and touchdown
- **Flight Cycles (FC)**: Complete take-off and landing sequences

When demand exceeds regular capacity, airlines face a critical trade-off:

- **Schedule maintenance too early**: Waste available aircraft utilization, reduce revenue-generating flight hours.
- **Schedule maintenance too late**: Risk capacity shortages when multiple aircraft become due simultaneously, creating hangar congestion.

When congestion occurs and regular capacity is insufficient, airlines must create additional maintenance slots through one of the following expensive options:

- **Overtime work**: Pay existing technicians 1.5-2× regular wages to work extra hours.
- **Subcontracting**: Hire third-party MRO (Maintenance, Repair, Overhaul) providers at 2-3× normal cost
- **Ground aircraft**: Leave planes waiting for available capacity, losing $75K-$120K of commercial revenue per day[2].

Airlines size their maintenance facilities (hangars, technicians, equipment) for normal demand patterns. However, stochastic events can cause sudden congestion. E.g., multiple aircrafts may simultaneously reach their maintenance limits due to higher-than-expected utilization, or ongoing checks may take longer than planned due to faults discovered. This disrupts planned maintenance schedules and increase operational costs. While larger permanent capacity would avoid these expenses, it is prohibitively expensive for occasional surges.

The Aircraft Maintenance Check Scheduling (AMCS) problem addresses the optimization of when and how often maintenance checks should be performed on the aircrafts in a fleet. Airlines must schedule A-checks (light maintenance every 2-3 months), C-checks (heavy maintenance every 18-24 months), and D-checks (major overhaul every 6-10 years) while considering uncertainty in aircraft daily utilization and maintenance elapsed time. The primary goal is to avoid being overly conservative and wastefully scheduling checks while the secondary

optimization goal (optional under this scope) is to minimize the frequency and number of these expensive additional slots through better scheduling

Problem Statement

For this problem, we are considering two periods, 2017-2020 & 2019-2022, over which multiple A-type and C-type checks are to be performed for a given airline fleet. The solution aims to minimize total wasted flight hours between maintenance checks while reducing the need for additional costly maintenance slots, increasing aircraft availability and reducing operational costs.*

**Note: Researchers can consider a period of 6 months starting from the start date mentioned in the dataset shared instead*

Input

- A fleet of 40-50 aircrafts (Airbus A320 family)
- Hangar capacity (To be determined based on dataset provided)
- Maintenance check intervals defined by thresholds (see Table 1)
- Stochastic aircraft utilization data (daily FH and FC follow distributions)
- Stochastic maintenance elapsed time data (checks take varying durations due to non-routine work discovered during inspection)

Primary Objective

Minimize total unused flight hours (FH) across all maintenance checks, which prevents overly conservative scheduling and maximizes aircraft availability for commercial operations.

Secondary Objective (Optional)

Minimize the total number of extra maintenance slots created over the planning horizon.

Constraints

- **Airworthiness:** No aircraft can exceed maximum utilization thresholds (DY, FH, or FC). If scheduling within regular capacity would cause a threshold violation, an extra maintenance slot must be created (with associated cost penalty $\lambda = 10^5$).
- **Hangar Capacity:** Maintenance slot assignment should not exceed the Hangar capacity per day per check type (e.g., max 2 A-check slots, 3 C-check slots per day). A-checks can be merged into C-checks.
- **Maintenance Duration:** Maintenance slot assignments should be consecutive if a check requires multiple consecutive days
- **Maintenance Intervals:** The maintenance schedule should maintain the minimum days required between consecutive same-type checks (e.g., 3 days between C-checks due to shared equipment, crew scheduling, hangar reorganization)

***Table 1**: Maintenance Check Intervals (Airbus A320 family)*

Aircraft Type	A-Check Interval (DY / FH / FC)	C-Check Interval (DY / FH / FC)
A319	120 / 750 / 750	730 / 7500 / 5000
A320	120 / 750 / 750	730 / 7500 / 5000
A321-1	120 / 750 / 750	730 / 7500 / 5000
A321-2	120 / 750 / 750	1096 / 12000 / 8000

Additional Information

- The solution could be adapted to new real-time FH, FC, and elapsed maintenance time information for adjusting future schedules accordingly.

Dataset Description

The dataset is shared along with this requirement document. Following is a brief description of the data. It contains real maintenance scheduling data from a major European airline operating an Airbus A320 fleet. The dataset includes the following three files:

- Check Scheduling Input (2017-2020): Data input for a period from 2017-2020
- Check Scheduling Input (2019-2022): Data input for a period from 2019-2022
- README.txt: Data dictionary

Solution Approach

Researchers are free to explore various optimization methodologies to address this problem including quantum enhanced optimization techniques. Data contains stochasticity which can be incorporated into your optimization approach. Following are the sources of variability:

Aircraft Daily Utilization

- Daily flight hours (FH) and flight cycles (FC) for each aircraft follow normal distributions
- Mean values can be estimated from historical data
- Actual values only known at end of each day
- Cumulative utilization determines when next maintenance check is due

Maintenance Check Elapsed Time

- Expected duration can be estimated from historical check data
- Actual duration known when check starts
- Extra days needed for maintenance follow empirical distributions based on non-routine work
- Example: C-check may take -2, -1, 0, +1, or +2 extra days beyond mean duration
- A-checks typically take approximately 1 day with minor variability

Analysis of Results

Solution quality can be evaluated using the following Key Performance Indicators (KPIs)/ metrics:

- **Average Flight Hours per Check:** Higher is better (less wasted utilization). This ensures scheduling is not overly conservative.
- **Total Number of Maintenance Checks:** Fewer checks = higher efficiency. Correlated with average FH per check.
- **Aircraft Availability:** Percentage of time aircraft are available for commercial service. Higher is better.
- **Schedule Feasibility:** All checks accommodated within regular capacity or with minimal extra slots.